

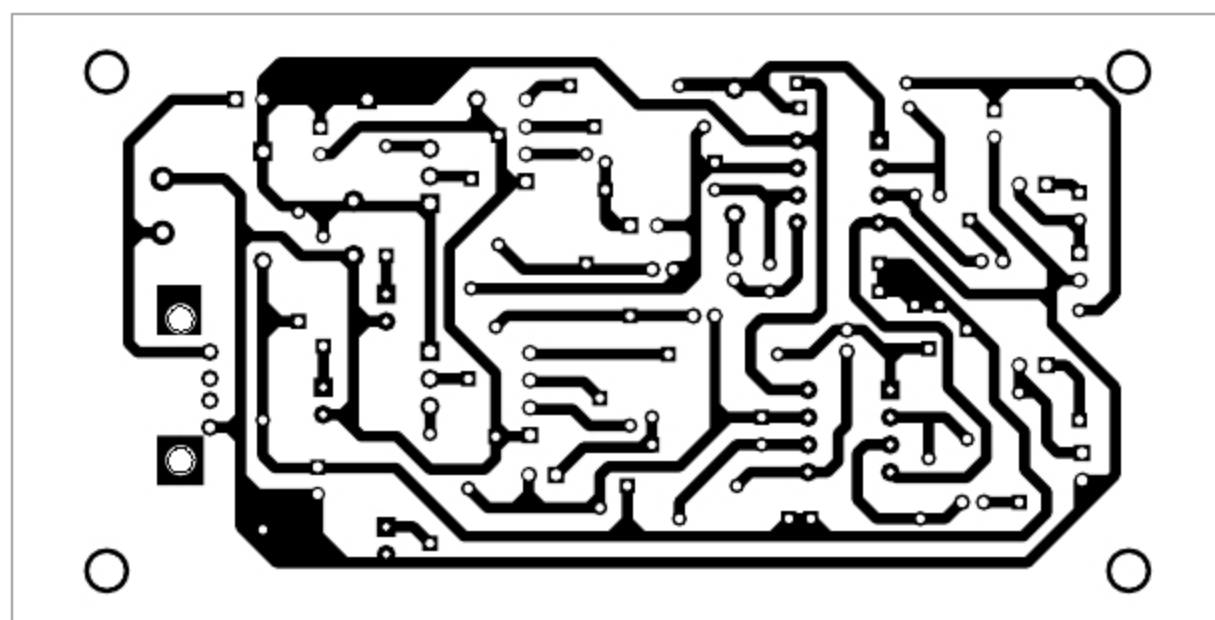


Fig. 2: Actual-size PCB layout for dual preamplifier circuit

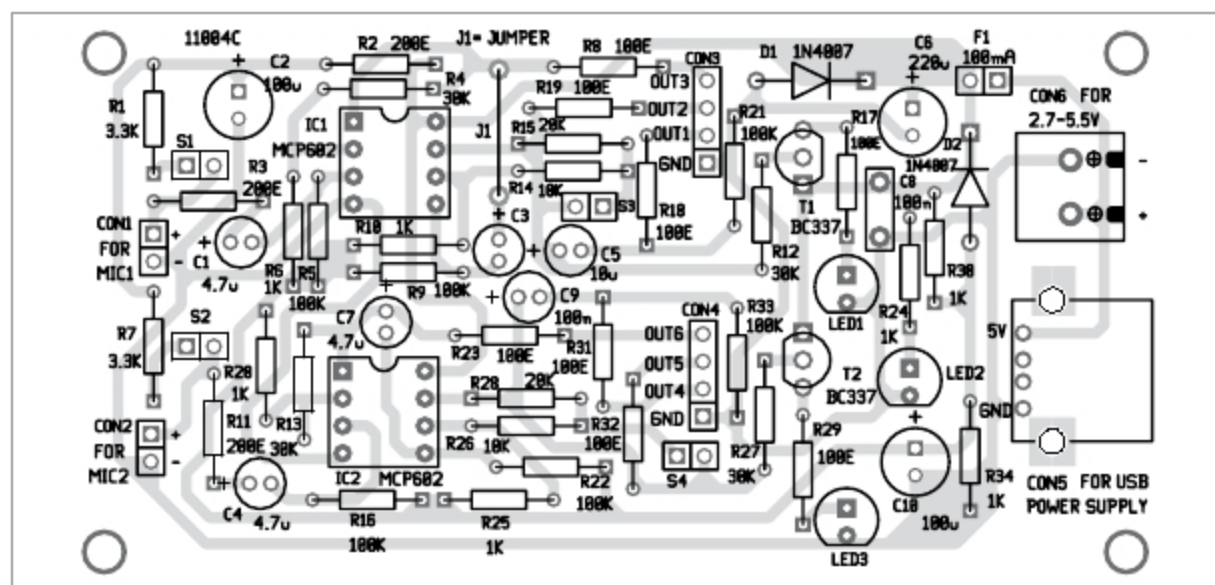


Fig. 3: Components layout of the PCB

and CON4, respectively. Preamplifier channels have six outputs—OUT1 through OUT6. While four outputs (OUT2, OUT3, OUT5, and OUT6) have DC components of around $V_{cc}/2$, two outputs (OUT1 and OUT4) do not have DC components. All outputs have series protecting resistors.

DC power supply (2.7V-5.5V) can be applied to connector CON6 or 5V supply on the USB connector CON5. Resistors R30 and R34, capacitor C10, and diode D2 produce half of the power supply needed for biasing the op-amp. Diode D1 and fuse F1 protect the circuit from reverse polarity con-

nection of the power supply.

Each audio channel has a buffer driving an LED to indicate the level of the output signal. LED1 and LED3 should be of low forward drop voltage—about 1.6V or below with low forward current. T1 and T2 drive LED1 and LED3, respectively.

Construction and testing

An actual-size PCB layout for dual microphone preamplifier is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, connect a 5V USB power supply across CON5. The glowing of

PARTS LIST

Semiconductors:

IC1, IC2	- MCP602 op-amp
T1, T2	- BC337 transistor
D1, D2	- 1N4007 rectifier diode
LED1-LED3	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R7	- 3.3-kilo-ohm
R2, R3, R11	- 200-ohm
R4, R12, R13, R27	- 30-kilo-ohm
R5, R9, R16, R21, R22, R33	- 100-kilo-ohm
R6, R10, R20, R24, R25, R30, R34	- 1-kilo-ohm
R8, R17-R19, R23, R29, R31, R32	- 100-ohm
R14, R26	- 10-kilo-ohm
R15, R28	- 20-kilo-ohm

Capacitors:

C1, C3, C4, C7	- 4.7 μ F, 16V electrolytic
C2, C10	- 100 μ F, 16V electrolytic
C5, C9	- 10 μ F, 16V electrolytic
C8	- 100nF ceramic disk
C6	- 220 μ F, 16V electrolytic

Miscellaneous:

CON1, CON2	- 2-pin connector
CON3, CON4	- 4-pin connector
CON5	- USB Type A female connector
CON6	- 2-pin terminal connector
MIC1, MIC2	- Electret microphone
S1-S4	- On/off switch
	- 8-pin IC base (2 no)
	- 5V regulated/USB power supply

LED2 indicates the presence of power in the circuit. The outputs (CON3 and CON4) can be directly connected to an ADC, MCU kit, or Arduino.

The circuit does not need any adjustment to work properly. If required, the gain of the preamplifier can be optimised for particular microphones or required output levels of the signals. **EFY**

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